

Superposition: More Features than Dimensions

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Meaning as direction in embedding space ([Note DL-072](#)).
- **Compare with:** Curse of dimensionality ([Note ML-085](#)).

1. Overview

Key point: A space of d dimensions holds at most d exactly perpendicular directions, but far more **nearly** perpendicular ones. Two random directions in 12,288 dimensions are within 5 degrees of perpendicular for every one of 20,000 pairs we drew, and the number of nearly perpendicular directions that fit grows exponentially with the dimension (the Johnson–Lindenstrauss lemma). The **superposition hypothesis** (G-1916) says that neural networks use this room: they store more features than they have dimensions, as nearly perpendicular directions, and accept a little interference between them. In a toy model, 5 features squeezed into 2 numbers are all stored, as a pentagon, but only when each feature is rarely active.

The [meaning as direction Note](#) showed that an embedding stores ideas such as “female” or “plural” as directions. The [MLP stores facts Note](#) showed that the feed-forward block reads directions with the rows of its first matrix and writes directions with the columns of its second. Both raise a question: how many such directions can a vector of 768 or 12,288 numbers hold without the ideas getting mixed up?

This Note answers it in four steps:

1. Random directions in high dimensions are almost always nearly perpendicular (section 4).
2. There is a hard limit on how perpendicular many directions can be, and a generous count of how many fit (sections 5 and 6).
3. The token vectors of a real transformer, GPT-2 small, spread out in the same way (section 7).
4. A small trained model shows superposition happening, and when it does not (section 8).

Figure 1 shows the first fact. Watch the spread of angles shrink towards 90 degrees as the dimension grows.

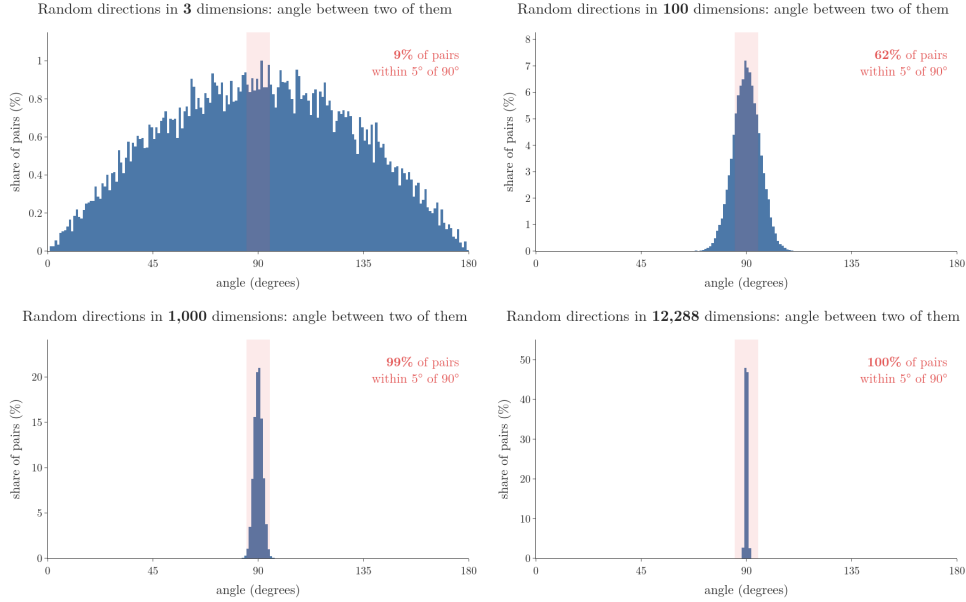


Figure 1: Angles between 20,000 random pairs of directions, as the dimension grows from 2 to 12,288. The red band marks 85 to 95 degrees

Extra: Two ideas of this Note follow Sanderson’s “How might LLMs store facts” (3Blue1Brown, Deep Learning Chapter 7, 2024): that a space with room for only n perpendicular directions has room for far more nearly perpendicular ones, so it can hold more ideas than it has dimensions (18:03–20:43); and the experiment of nudging 10,000 random 100-dimensional vectors to be more perpendicular while watching the distribution of the angles between them (18:34–19:37). Our figures, the dimension sweep, the bound of section 5 and the toy model are our own; section 5 shows what the optimisation can and cannot reach.

2. Prerequisites

- [MLP stores facts Note](#): rows of the first matrix read directions, columns of the second write directions; a neuron is one hidden value of the MLP.
- [Meaning as direction Note](#): ideas stored as directions; a dot product with a direction reads how much of the idea a vector carries.
- [Dot product and cosine similarity Note](#): perpendicular vectors have dot product 0; the cosine of the angle between two vectors.
- [Curse of dimensionality Note](#): high-dimensional spaces behave very differently from the 2-D and 3-D spaces we can picture.
- [Activation functions Note](#) (ReLU) and [Adam Note](#), for the toy model of section 8.

3. Perpendicular directions do not interfere, but only d of them fit

Key point: If each feature is a direction and we read a feature with a dot product, every other active feature leaks into the reading in proportion to the cosine between their directions. Perpendicular directions leak nothing, but a d -dimensional space has room for only d of them.

Suppose a vector h stores several features, each as a unit direction f_i scaled by how strongly the feature is present, x_i :

$$h = x_1 f_1 + x_2 f_2 + \cdots + x_n f_n$$

To read feature j back, take the dot product with its direction:

$$h \cdot f_j = x_j + \sum_{i \neq j} x_i (f_i \cdot f_j)$$

The first term is what we want. The sum is **interference** (G-962): every other active feature leaks in, weighted by the cosine $f_i \cdot f_j$. If all the directions are perpendicular, every cosine is 0 and the reading is exact (Figure 2).

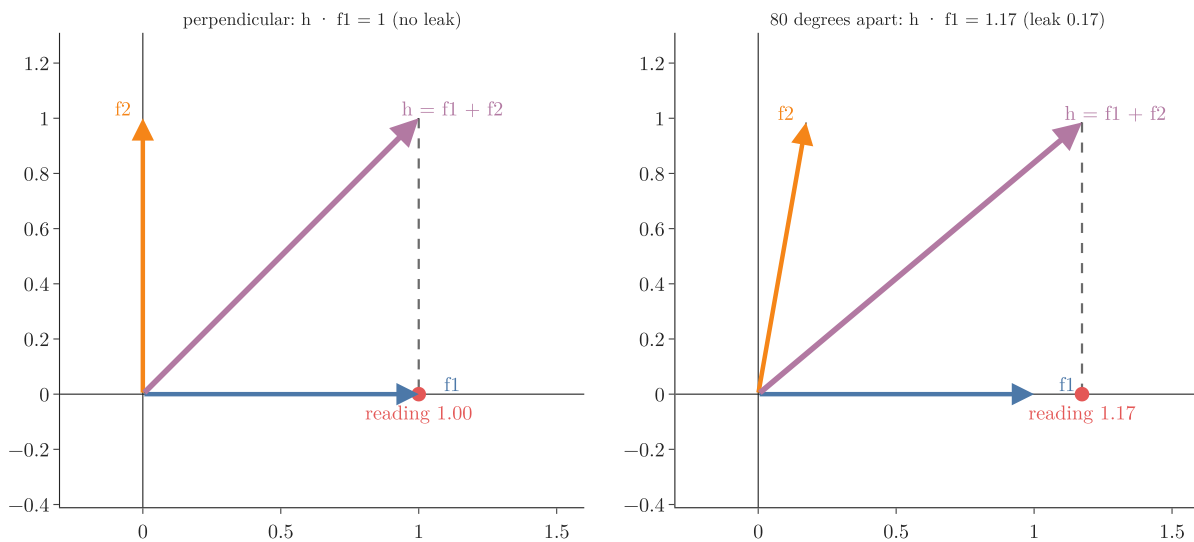


Figure 2: An example of our own: two features, both with strength 1, so $h = f_1 + f_2$. Reading feature 1 with $h \cdot f_1$ projects h onto f_1 (dashed). Left: perpendicular directions give exactly 1. Right: directions 80 degrees apart give $1 + \cos 80^\circ = 1.17$; the extra 0.17 is the leak from feature 2

The catch: non-zero vectors that are all perpendicular to each other are **linearly independent** (G-1102), and a d -dimensional space has at most d independent vectors. So with exact

perpendicularity, d dimensions store at most d features. GPT-2 small’s vectors have $d = 768$ numbers. To store more than 768 features in them, the directions must be allowed to be only **nearly perpendicular** (G-1307), at the price of a little interference.

4. Random directions are nearly perpendicular

Key point: Two random directions in d dimensions have a cosine with mean 0 and mean square $1/d$, so their angle is spread about $57.3/\sqrt{d}$ degrees around 90. In 100 dimensions, 62 percent of random pairs lie within 5 degrees of perpendicular; in 1,000 dimensions, 99 percent.

1. **In words:** fix one unit vector v and pick another, u , uniformly at random on the sphere. By symmetry no coordinate of u is special, so each u_k^2 has the same average; the d of them add up to $\|u\|^2 = 1$, so each averages $1/d$.
2. **Formula:** the cross terms $u_k u_l$ average to zero (flipping the sign of u_k is equally likely), so

$$E[(u \cdot v)^2] = \sum_k v_k^2 E[u_k^2] = \frac{1}{d} \sum_k v_k^2 = \frac{1}{d}$$

The cosine has mean 0 and a typical size of $1/\sqrt{d}$. Near 90 degrees a small cosine c is an angle about c radians away from 90 degrees, that is $57.3c$ degrees, so the angle is spread about $57.3/\sqrt{d}$ degrees.

3. **Example:** $d = 100$ gives $57.3/10 = 5.7$ degrees.

The Notebook draws 20,000 random pairs for each dimension (Figure 1):

Dimension d	Mean of $\cos^2$ (theory $1/d$)	Spread of the angle, degrees (theory $57.3/\sqrt{d}$)	Pairs within 5 degrees of 90
3	0.331 (0.333)	39.0 (33.1)	9 percent
10	0.0996 (0.1)	19.0 (18.1)	20 percent
100	0.0100 (0.01)	5.75 (5.73)	62 percent
1,000	0.0010 (0.001)	1.82 (1.81)	99 percent
12,288	0.0001 (0.0001)	0.52 (0.52)	100 percent

The mean square matches $1/d$ in every row. The spread matches $57.3/\sqrt{d}$ once angles stay close to 90 degrees ($d \geq 30$); in 3 dimensions the small-angle step does not hold (Figure 3).

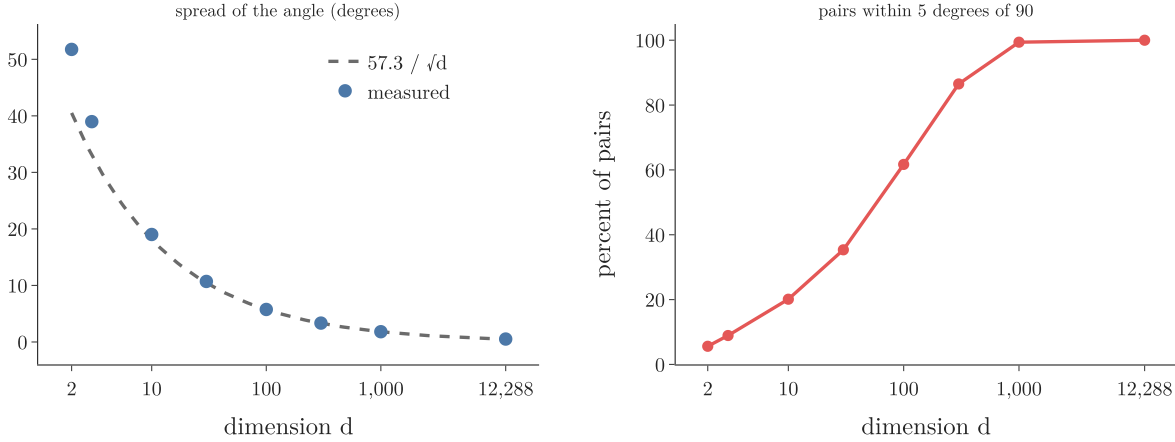


Figure 3: Left: the measured spread of the angle between random directions (dots) against the prediction $57.3/\sqrt{d}$ (dashed). Right: the share of pairs within 5 degrees of perpendicular rises from 6 percent in 2 dimensions to 100 percent in 12,288

In 2 or 3 dimensions random directions point anywhere; in thousands of dimensions they are almost all nearly perpendicular. This is one more way in which high-dimensional space is unlike the space we can picture (the [curse of dimensionality Note](#)).

5. How perpendicular can many directions be made?

Key point: For N unit directions in d dimensions the mean squared cosine is at least $(N/d - 1)/(N - 1)$. For 10,000 directions in 100 dimensions that floor is 0.0099, almost exactly what random directions already have (0.0100). Optimisation can still trim the worst pairs, from 30.9 to 15.8 degrees away from perpendicular, but it cannot narrow the bulk of the histogram.

Random directions are a starting point. Can we nudge them to be *more* perpendicular? First, a limit.

1. **In words:** put the N unit vectors as the rows of a matrix U ($N \times d$). The matrix of all their dot products, $G = UU^T$, has 1s on its diagonal and the cosines elsewhere. G is built from a d -column matrix, so it has at most d non-zero eigenvalues, and those eigenvalues must add up to the diagonal sum, N . Numbers with a fixed sum have the smallest sum of squares when they are equal.
2. **Formula:** with eigenvalues $\lambda_1, \dots, \lambda_d$ adding up to N ,

$$\sum_{i,j} G_{ij}^2 = \sum_k \lambda_k^2 \geq \frac{N^2}{d}$$

The diagonal contributes N , so the $N(N - 1)$ off-diagonal cosines satisfy

$$\text{mean of } \cos^2 \geq \frac{N^2/d - N}{N(N - 1)} = \frac{N/d - 1}{N - 1}$$

3. **Example:** $N = 10,000$, $d = 100$: the mean of $\cos^2$ is at least $99/9,999 = 0.0099$, a root-mean-square cosine of at least 0.0995. Since the mean square is at least that big, some pair must have a cosine of 0.0995 or more: at least 5.7 degrees away from perpendicular. No arrangement keeps every pair within 89–91 degrees, which would need every cosine below 0.0175.

Random directions have a mean of $\cos^2$ equal to $1/d = 0.0100$, a hair above the floor of 0.0099. So the bulk of the histogram cannot get narrower. What optimisation *can* do is pull in the rare pairs far from 90 degrees.

The Notebook takes 10,000 random directions in 100 dimensions and runs 60 steps of **gradient descent** (G-862) on the sum of $\cos^8$ over all pairs (the high power pushes hardest on the worst pairs), rescaling each vector to length 1 after every step. Figure 4 shows the histogram at every step, on a log scale so that the rare far-off pairs stay visible.

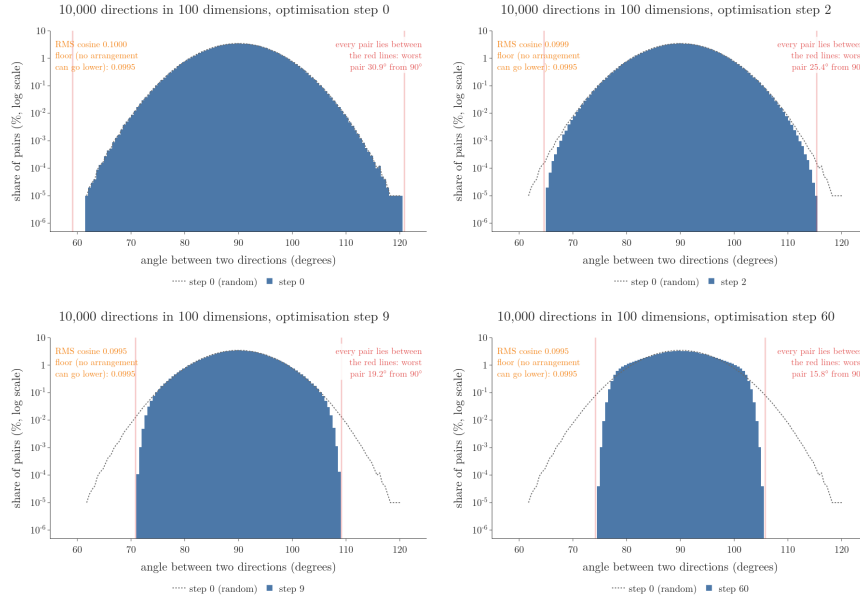


Figure 4: 10,000 directions in 100 dimensions, nudged to be more perpendicular. Dotted grey: the random start. Red lines: every pair lies between them. The log scale shows the rare pairs far from 90 degrees

Step	Worst pair, degrees from 90	Root-mean-square cosine	Pairs within 89–91 degrees
0 (random)	30.9	0.1000	13.8 percent
10	18.8	0.0995	13.7 percent
60	15.8	0.0995	13.1 percent

Watch the red lines move in while the middle of the histogram stays put. The root-mean-square cosine drops to the floor, 0.0995, within 10 steps and cannot go lower; the share of pairs within 89–91 degrees stays near 13 percent.

6. The room grows exponentially with the dimension

Key point: The Johnson–Lindenstrauss lemma implies that the dimension needed for n nearly perpendicular directions grows only with $\log n$. Turned around, the number of directions that fit grows exponentially with the dimension: going from about 5,900 to 11,800 dimensions raises the guaranteed count from a thousand to a million.

The **Johnson–Lindenstrauss lemma** (G-984), in the form proved by Dasgupta and Gupta (2003, Theorem 2.1): for any $0 < \varepsilon < 1$ and any n points in $\mathbb{R}^D$, if

$$k \geq \frac{4 \ln n}{\varepsilon^2/2 - \varepsilon^3/3}$$

then there is a map f into $\mathbb{R}^k$ that keeps every squared distance within a factor $1 \pm \varepsilon$:

$$(1 - \varepsilon)\|u - v\|^2 \leq \|f(u) - f(v)\|^2 \leq (1 + \varepsilon)\|u - v\|^2$$

Their map is a scaled projection onto a random k -dimensional subspace, so it is linear and sends the origin to the origin.

From distances to angles. Take the n perpendicular unit vectors $e_1, \dots, e_n$ of $\mathbb{R}^n$, plus the origin: $n + 1$ points.

1. Distances to the origin are kept, so each image has $\|f(e_i)\|^2$ between $1 - \varepsilon$ and $1 + \varepsilon$.
2. $\|e_i - e_j\|^2 = 2$, so $\|f(e_i) - f(e_j)\|^2$ lies between $2(1 - \varepsilon)$ and $2(1 + \varepsilon)$.
3. Expanding a squared distance, $f(e_i) \cdot f(e_j) = \frac{1}{2} (\|f(e_i)\|^2 + \|f(e_j)\|^2 - \|f(e_i) - f(e_j)\|^2)$, which therefore lies between -2ε and 2ε .
4. Dividing by the lengths, each cosine is at most $2\varepsilon/(1 - \varepsilon)$ in size.

So n directions with every cosine below $2\varepsilon/(1 - \varepsilon)$ fit in k dimensions. With $\varepsilon = 0.1$, every cosine is at most 0.222, every angle within 12.8 degrees of 90, and (Notebook):

Directions n	Dimension k that guarantees it
1,000	5,922
1,000,000	11,842
1,000,000,000	17,763
1,000,000,000,000	23,684

Each extra 5,920 or so dimensions multiplies the guaranteed number of directions by 1,000 (Figure 5).

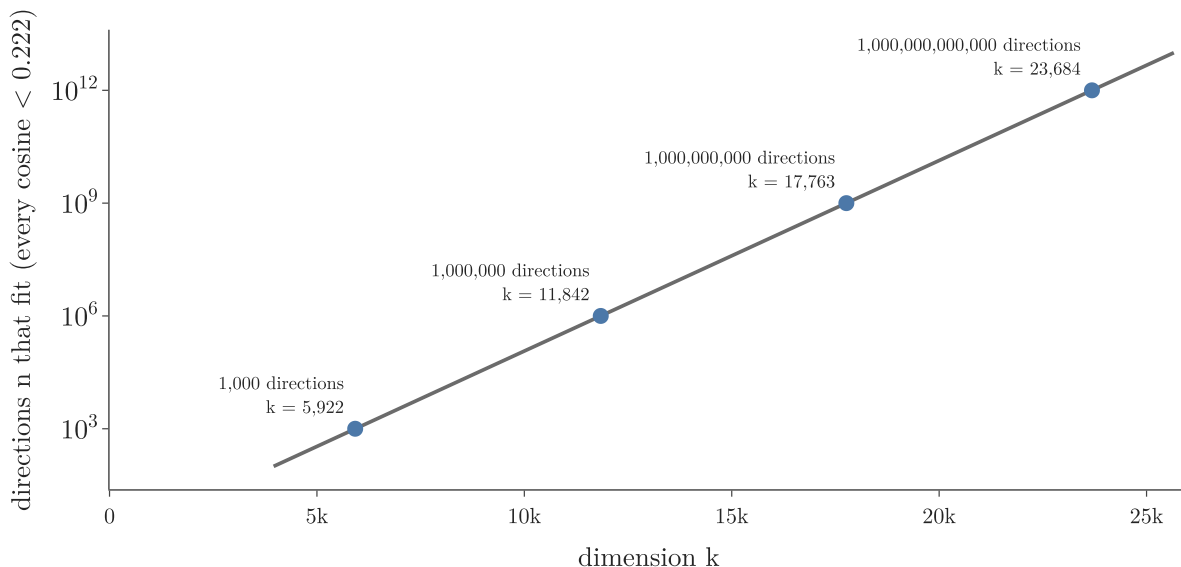


Figure 5: The Johnson–Lindenstrauss guarantee with $\varepsilon = 0.1$: the number of directions that fit, every cosine below 0.222, against the dimension. On a log scale the line is straight: the count grows exponentially with the dimension

The guarantee is cautious: in the Notebook, 10,000 *random* directions in 768 dimensions already have every pair within 12.1 degrees of perpendicular, and in 12,288 dimensions within 2.9 degrees. Elhage et al. (2022, “The Superposition Hypothesis”) point to the same lemma: only n orthogonal vectors fit in n dimensions, but $\exp(n)$ many “almost orthogonal” ones do.

7. Real token vectors spread out the same way

Key point: GPT-2 small stores 50,257 tokens in 768 dimensions, 65 times more tokens than dimensions. Apart from one shared mean vector, its token vectors sit at 90.0 degrees from each other on average, with a spread of 2.8 degrees, close to random directions (2.1 degrees).

The Notebook takes 3,000 random rows of GPT-2 small’s token-embedding table (the [meaning as direction Note](#), section 7) and measures all 4.5 million angles between them (Figure 6).

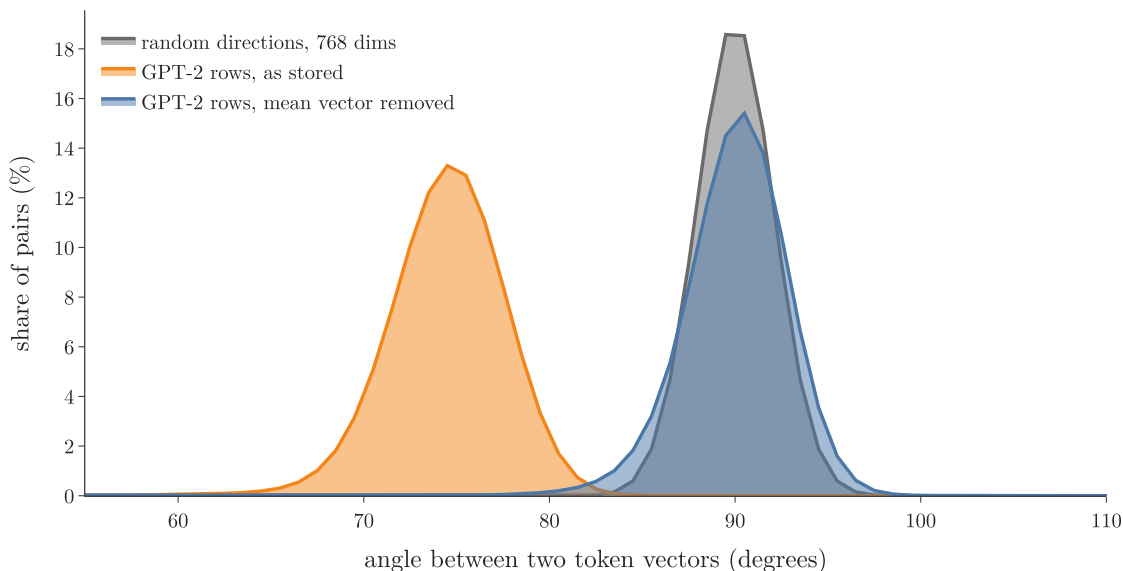


Figure 6: Angles between 3,000 token vectors of GPT-2 small. Orange: as stored. Blue: after subtracting the mean of all 50,257 token vectors. Grey: random directions in 768 dimensions

	Mean angle	Spread	Pairs within 5 degrees of 90
GPT-2 rows, as stored	74.4	3.2	0 percent
GPT-2 rows, mean vector removed	90.0	2.8	93 percent
Random directions, 768 dimensions	90.0	2.1	98 percent

As stored, every token vector shares a common component, the table’s mean vector, so all pairs lean towards each other (74 degrees). Remove that one shared direction, and the 50,257 token vectors spread out almost like random directions: 93 percent of pairs lie within 5 degrees of perpendicular, although there are 65 times more tokens than dimensions.

8. The superposition hypothesis and a toy model

Key point: The superposition hypothesis says networks represent more features than they have dimensions, as nearly perpendicular directions. In a toy model with 5 features and 2 hidden numbers, a ReLU model stores only the 2 most important features when features are always active, but all 5, as a pentagon, when each is active only 5 percent of the time. Without the ReLU it never stores more than 2.

The hypothesis. Elhage et al. (2022) name the idea: “the superposition hypothesis”. Neural networks “want to represent more features than they have neurons”, so they exploit the room of section 6 to simulate a larger network. A **feature** (G-772) here is a property of the input the network represents, such as “this token refers to a specific famous person” or “this clause describes music” (the paper’s examples). Superposition is offered as a *hypothesis*:

it explains observations, and toy models show it can happen, but it is not proven for large models. It explains one observation in particular: many neurons are **polysemantic** (G-1517), responding to several unrelated features.

8.1 The toy model

Key point: Squeeze 5 features into 2 numbers with a matrix W , unsqueeze with W^T , add a bias and apply ReLU. Training decides which features get a direction.

The setup follows Elhage et al. (2022, “Demonstrating Superposition”):

- **Data:** each observation has 5 features $x_1, \dots, x_5$. Each feature is 0 with probability S , the **sparsity**, and otherwise a random number between 0 and 1.
- **Model:** $h = Wx$ squeezes the 5 numbers into 2 (W is 2×5). The output is $x' = \text{ReLU}(W^T h + b) = \text{ReLU}(W^T W x + b)$. Column W_i is feature i ’s direction in the 2-number space.
- **Loss:** the weighted squared error $\sum_i I_i (x_i - x'_i)^2$, where the **importance** (G-925) I_i says how much each feature matters.
- **Training (Notebook):** Adam, 6,000 steps of 1,024 random observations, 5 random starts per setting; we keep the run with the lowest loss (for its 10-feature, 5-neuron figures the paper keeps the best of 1,000 runs).

8.2 Sparse features are stored in superposition

Key point: As the features get sparser, the model stores 2, then 4, then all 5 of them in its 2 numbers.

Figure 7 shows the trained directions W_i with importance falling as $I_i = 0.7^i$ (feature 1 matters most). A feature counts as stored when its direction has length above 0.5.

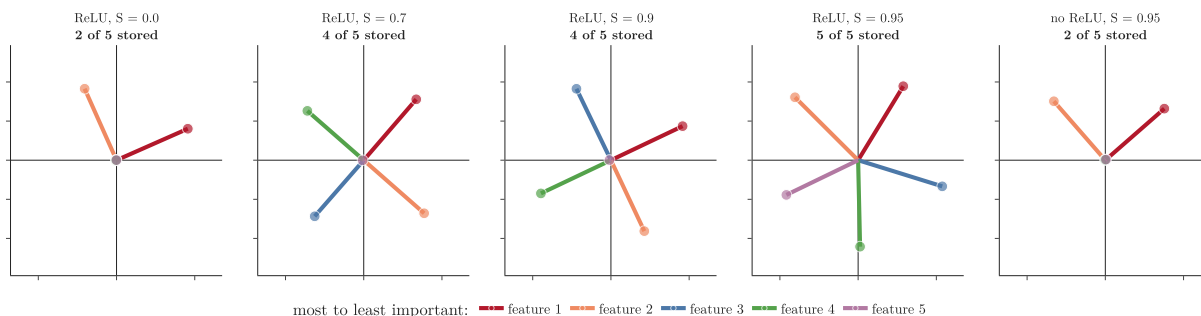


Figure 7: Toy model of superposition: the 2-number direction of each of 5 features after training, as features get sparser. Last panel: no ReLU

Model	Sparsity S	Features stored	What the directions look like
ReLU	0 (always active)	2	features 1 and 2 at 90 degrees; 3, 4, 5 dropped
ReLU	0.7	4	two opposite pairs
ReLU	0.9	4	two opposite pairs
ReLU	0.95	5	all five, spread around the circle
no ReLU	0.95	2	features 1 and 2 at 90 degrees

With dense features the model does what a linear method such as PCA would do: it keeps the two most important features on perpendicular directions and drops the rest. Elhage et al. (2022) describe the same result: “With dense features, the model learns to represent an orthogonal basis of the most important two features”. As features get sparser, more of them share the 2 numbers. Without the ReLU, the model keeps only 2 even when features are sparse: “the linear model always learns the top m most important features” (Elhage et al. 2022).

With 5 equally important features ($I_i = 1$) at $S = 0.95$, the directions spread out evenly. Figure 8 shows the training. Watch the 5 arrows push apart until the angles between neighbours are all close to 72 degrees, the angles of a regular pentagon. The paper reports the same regular pentagon for this setting (Elhage et al. 2022, “Perturbing a Single Feature”).

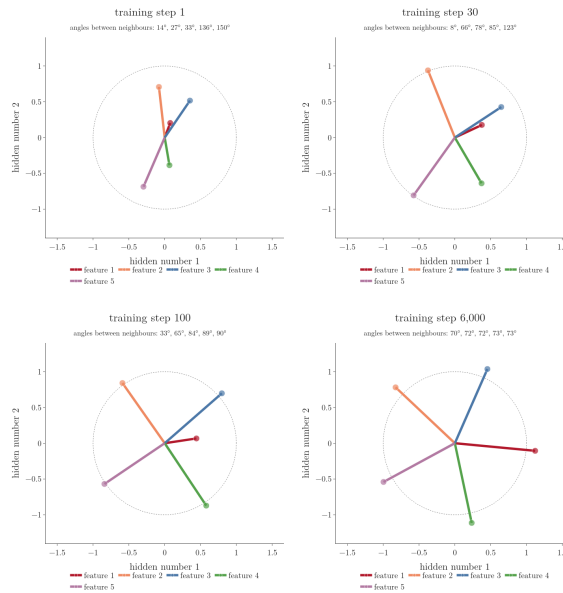


Figure 8: Training the toy model with 5 equally important features, each active 5 percent of the time, in 2 hidden numbers. The 5 directions spread into a pentagon

8.3 Why sparsity and the ReLU make it work

Key point: In the pentagon, one active feature leaks a little into its two near neighbours and a lot of negative into the two far ones; the negative bias and the ReLU cut most of it away. When two features are active at once, the leaks add up and the outputs go wrong, so superposition only pays when features are rarely active together.

From the trained pentagon (Notebook): every bias is about -0.24 , and the cosine-like products $W_i \cdot W_j$ are about 1.28 for a feature with itself, 0.39 for its two neighbours, and -1.04 for the two far features. Feed in feature 1 alone, $x = (1, 0, 0, 0, 0)$:

Output for feature	1	2 (far)	3 (near)	4 (near)	5 (far)
$W^T W x + b$	1.02	-1.26	0.16	0.14	-1.29
After ReLU	1.02	0	0.16	0.14	0

Feature 1 is read back almost exactly. The interference into the far features is negative, and the ReLU removes it entirely; the interference into the near features is mostly cancelled by the negative bias. Elhage et al. (2022) summarise this: superposition works “at the cost of ‘interference’ that requires nonlinear filtering”. A linear output has no ReLU to remove the negative leaks, which is why the no-ReLU model never stores more than 2.

Now make features 1 and 2 active together, $x = (1, 1, 0, 0, 0)$: the outputs become 0.01, 0.04, 0.59, 0, 0. Both active features are nearly lost, and feature 3, which is absent, reads 0.59. With $S = 0.95$ two features are active together only rarely, so the model accepts these rare errors in exchange for storing all 5. With dense features they would happen all the time, which is why the dense model stores only 2.

8.4 What this means for neurons

Key point: If a model stores features in superposition, a feature is a direction across many neurons, not one neuron, and one neuron takes part in many features.

The MLP of a transformer has 4 times as many hidden neurons as the vector has numbers (the [MLP stores facts Note](#)). Each neuron’s activation measures how well the vector aligns with one row of the first matrix. If the model stores more features than it has neurons, as nearly perpendicular directions, then a feature cannot be one neuron lighting up: it is a particular combination of neurons, and each neuron is shared by several features. That is the polysemantic behaviour the superposition hypothesis sets out to explain (Elhage et al. 2022). It also makes models hard to read neuron by neuron.

9. Summary

Fact	Evidence
Only d exactly perpendicular directions fit in d dimensions	perpendicular non-zero vectors are linearly independent
Random directions are nearly perpendicular; spread about $57.3/\sqrt{d}$ degrees	derivation; 62 percent within 5 degrees at $d = 100$, 99 percent at $d = 1,000$
Mean squared cosine of N directions is at least $(N/d - 1)/(N - 1)$	derivation; optimising 10,000 directions in 100-d trims the worst pair from 30.9 to 15.8 degrees but stays at the floor
The number of nearly perpendicular directions grows exponentially with d	Johnson–Lindenstrauss (Dasgupta and Gupta 2003); a million directions with every cosine below 0.222 in 11,842 dimensions
Real token vectors spread like random ones	GPT-2 small, mean vector removed: 90.0 ± 2.8 degrees
Sparse features are stored in superposition	toy model: 2 of 5 features stored when dense, all 5 (a pentagon) at 95 percent sparsity; never more than 2 without ReLU

- Perpendicular directions never interfere, but only d of them fit.
- Nearly perpendicular directions interfere a little, and exponentially many of them fit.
- The superposition hypothesis: networks store more features than dimensions this way, filtering the interference with non-linearities such as ReLU.
- It pays only when features are sparse, so that few of them are active at once.

10. Sources

Built from

- Sanderson, G. (3Blue1Brown), “How might LLMs store facts | Deep Learning Chapter 7”, 2024, 3blue1brown.com/lessons/mlp, <https://www.youtube.com/watch?v=9-Jl0dxWQs8>. Superposition as a hypothesis, nearly perpendicular directions, the distribution of angles between 10,000 optimised 100-dimensional vectors, and Johnson–Lindenstrauss (16:59–21:14).
- Elhage, N., Hume, T., Olsson, C., Schiefer, N., Henighan, T., Kravec, S., Hatfield-Dodds, Z., Lasenby, R., Drain, D., Chen, C., Grosse, R., McCandlish, S., Kaplan, J., Amodei, D., Wattenberg, M. and Olah, C. (2022). Toy Models of Superposition. *Transformer Circuits Thread*, 14 September 2022. transformer-circuits.pub/2022/toy_model. Sections “Background and Motivation”, subsection “The Superposition Hypothesis” (the hypothesis, polysemantic neurons, $\exp(n)$ almost-orthogonal vectors via Johnson–Lindenstrauss); “Demonstrating Superposition” (the data: sparsity S_i and importance I_i ; the linear and ReLU output models $x' = \text{ReLU}(W^T W x + b)$; the dense two-feature result; the linear model keeps the top m features; best of 1,000 runs); “The Geometry of Superposition”, “Perturbing a Single Feature” ($n = 5$, $m = 2$, $I = 1$, $1 - S = 0.05$ gives a regular pentagon); introduction (“interference that requires nonlinear filtering”).

Other references

- Dasgupta, S. and Gupta, A. (2003). An Elementary Proof of a Theorem of Johnson and Lindenstrauss. *Random Structures and Algorithms*, 22(1), 60–65. Theorem 2.1 (the bound on k and the $1 \pm \varepsilon$ distortion); proof of Theorem 2.1 (the map is $\sqrt{d/k}$ times the projection onto a random k -dimensional subspace).
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D. and Sutskever, I. (2019). Language Models are Unsupervised Multitask Learners. OpenAI. Weights: `openai-community/gpt2` (token-embedding table, $50,257 \times 768$), `huggingface.co`.

11. Key terms

Term	Meaning
Feature	A property of the input that a network represents, stored as a direction in a vector
Interference	The part of a feature’s reading that comes from other active features, in proportion to the cosines between their directions
Nearly perpendicular	Two directions whose angle is close to 90 degrees, so their cosine is close to 0
Johnson–Lindenstrauss lemma	n points can be mapped into about $\log n/\varepsilon^2$ dimensions with every squared distance kept within a factor $1 \pm \varepsilon$
Superposition	Storing more features than there are dimensions, as nearly perpendicular directions
Superposition hypothesis	The proposal that real neural networks store features in superposition (Elhage et al. 2022)
Polysemantic neuron	A neuron that responds to several unrelated features
Sparsity S	The probability that a feature is 0 in an observation; high sparsity means the feature is rarely active
Importance I_i	The weight of feature i in the toy model’s loss
Toy model of superposition	$x' = \text{ReLU}(W^T W x + b)$ with more features than hidden numbers, trained to rebuild its input